

AWS-BASED NETWORK AUDITING

Amazon VPC Flow Logs Analysis for Traffic Auditing

We turn raw VPC network activity into structured, queryable audit information.

HACKATHON DAY

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Why VPC traffic auditing needs visibility

01

Raw logs are hard to read

High-volume records arrive as dense, low-context text.

02

Investigations take too long

Manual filtering delays answers to basic traffic questions.

03

Audit trails are fragmented

Evidence is difficult to search, retain and present consistently.

04

Visibility gaps create risk

Teams cannot quickly explain who connected, where and when.

Turn network exhaust into audit evidence

COLLECT



Capture VPC Flow Logs reliably into durable cloud storage.

STRUCTURE



Parse and normalize key connection fields automatically.

QUERY



Make traffic searchable by source, destination, action and time.

PRESENT



Give auditors a clear, browser-based view of activity.

PROPOSED SOLUTION

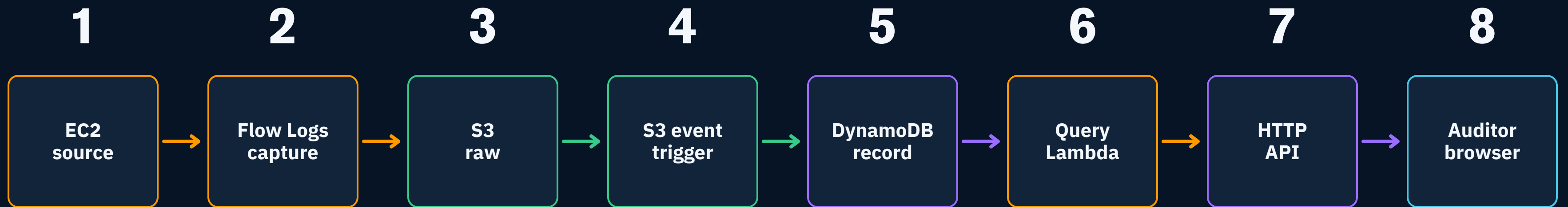
Follow one packet



One connection. One traceable path. One audit-ready answer.

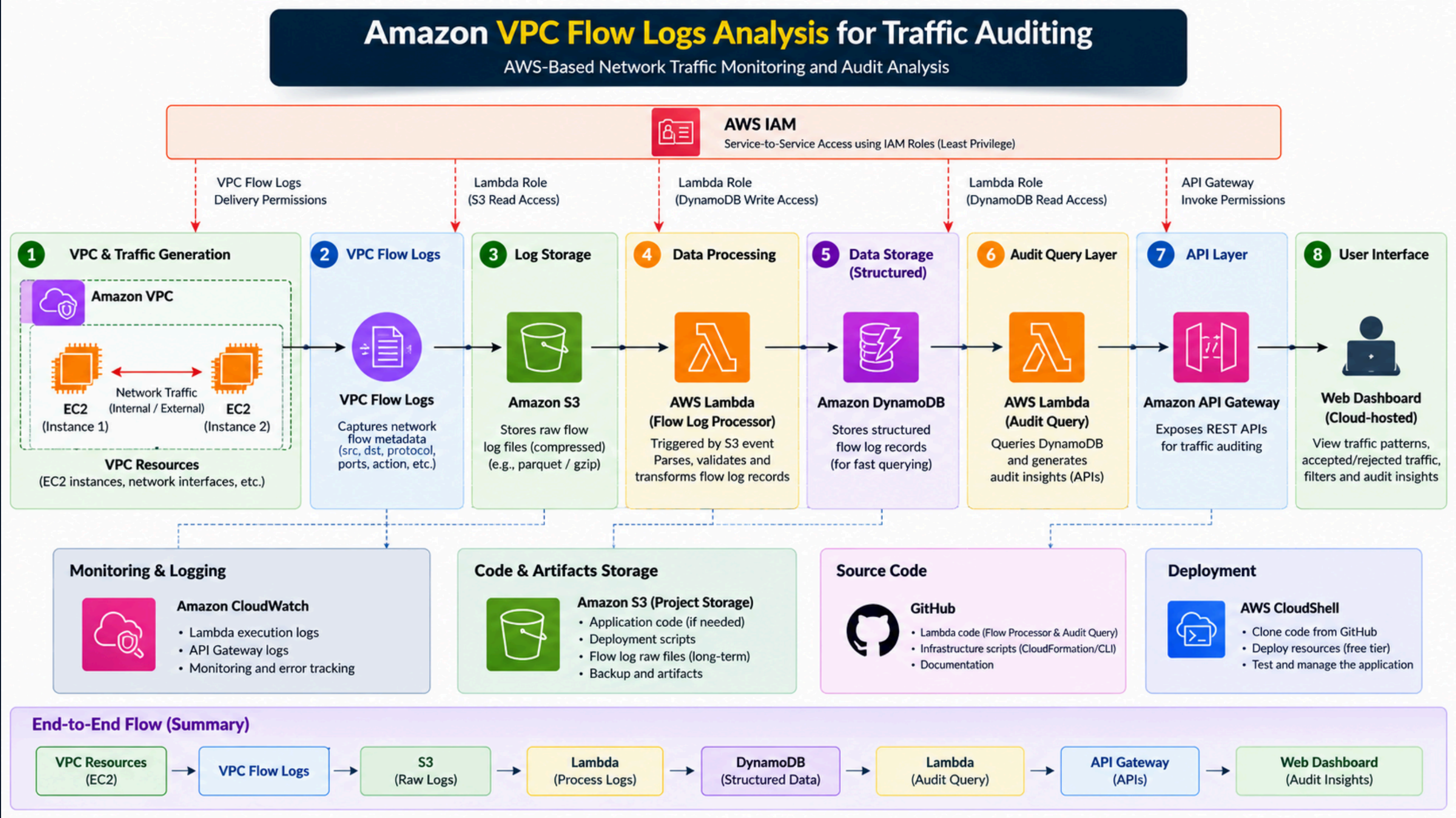
Follow one packet.

One connection. Eight observable hops. A complete audit story.



RAW SIGNAL → AUDIT-READY INSIGHT

End-to-end. Serverless. Event-driven.



Reserved for the final end-to-end architecture image — from VPC to dashboard.

A focused serverless stack

VPC FLOW LOGS

Capture connection metadata

S3

Durable raw-log landing zone

LAMBDA

Parse, transform and query

DYNAMODB

Store normalized audit records

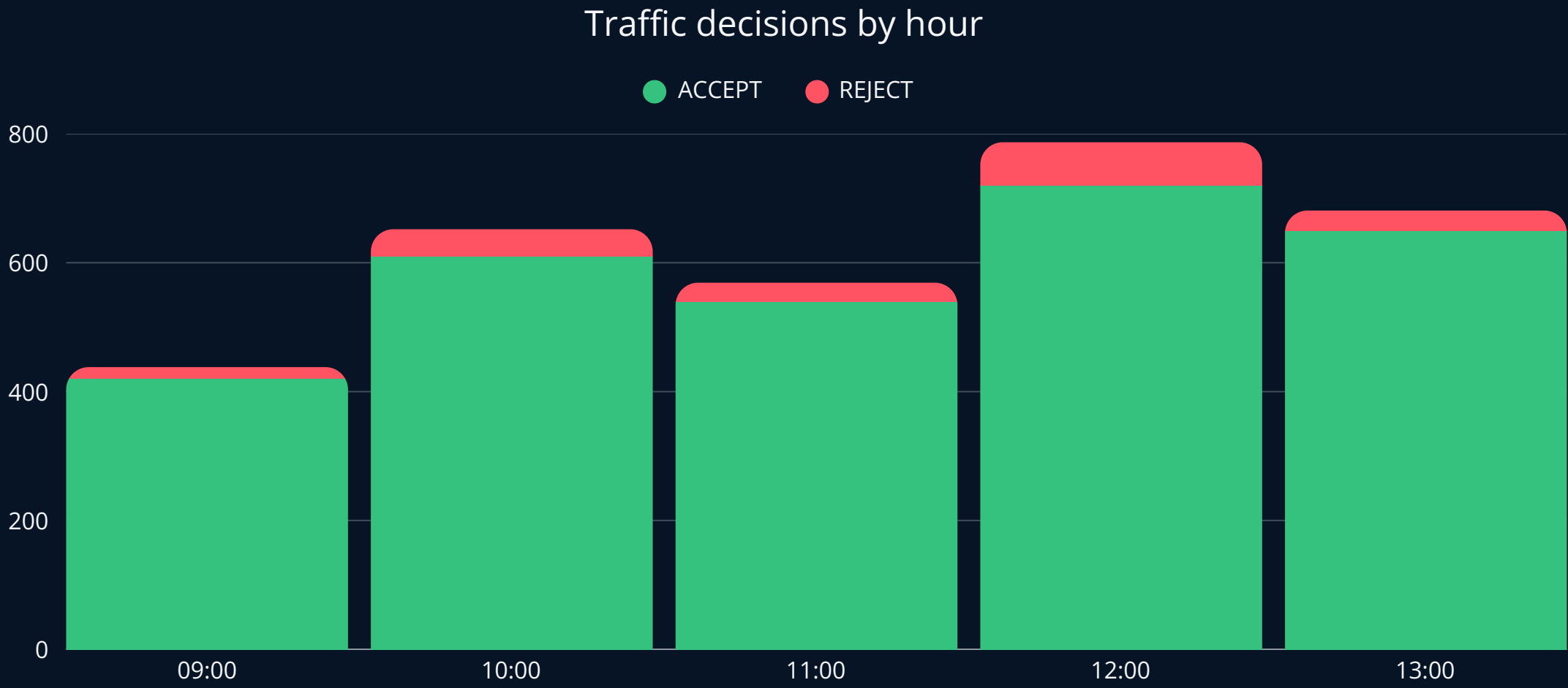
API GATEWAY

Expose controlled query endpoints

AUDIT DASHBOARD

Turn records into answers

What the auditor sees



FILTER

Source IP · Destination · Action · Time

AUDIT ANSWER

Fast, structured evidence for who connected, where and when.

DATA COLLECTION

VPC · Flow Logs · S3

DATA PROCESSING

S3 · Lambda · DynamoDB

DATA CONSUMPTION

DynamoDB · Lambda · API Gateway · Dashboard

Least privilege, split by responsibility

INGESTION LAMBDA ROLE

Process new flow logs

Read only from the designated S3 bucket
Write only normalized records to DynamoDB

**NO SHARED
SUPER-ROLE**

QUERY LAMBDA ROLE

Serve audit queries

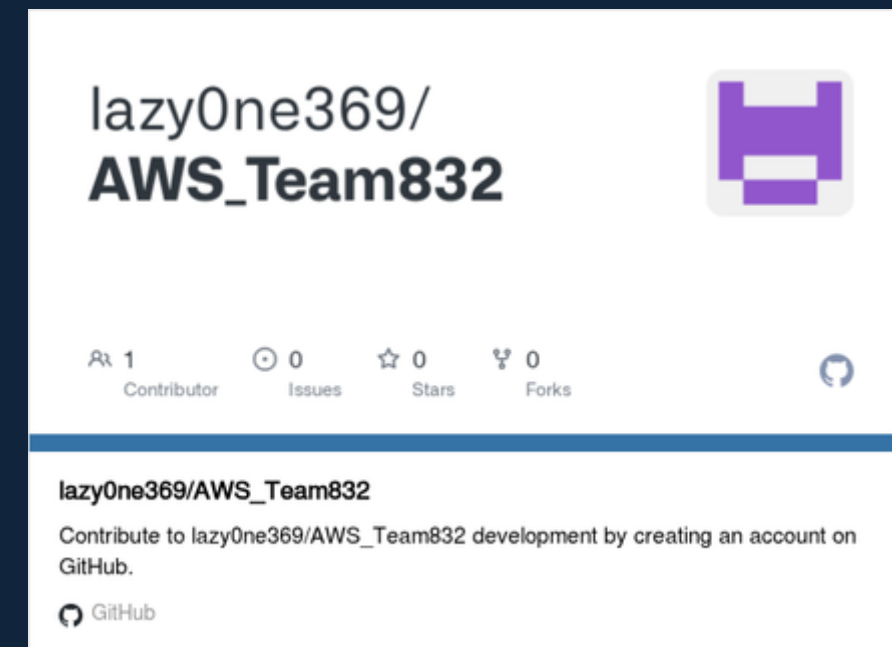
Read only from the audit table
Invoked only through the controlled API

100% cloud-deployed. Code on GitHub.

NOTHING RUNS LOCALLY

Ingestion, processing, storage, API and dashboard are deployed in the cloud. The repository contains the source and deployment artifacts.

GITHUB REPOSITORY



From connection to audit answer

- 01 Generate sample VPC traffic
- 02 Verify raw Flow Logs in S3
- 03 Watch Lambda normalize records
- 04 Query the live audit dashboard

CLICK TO OPEN LIVE DASHBOARD

[dashboard@link](#)

Visibility first. Smarter auditing next.

EXPECTED OUTCOMES

Queryable network activity
Faster audit investigations
Consistent cloud-native evidence
Clear visibility into ACCEPT / REJECT flows

FUTURE SCOPE

Richer filters and saved queries
Longer retention and export workflows
Cross-account audit views
Trend reporting for governance

Focus: visibility and auditing of network traffic — not automatic threat detection.

Raw network activity, turned into audit-ready insight.

THANK YOU